

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	Horizontal – constant speed (with or without reference to force) (1) Vertical – [constant] acceleration due to force of gravity [acting downwards] (1)	2			2		
		(ii)	Similarity- projectile will [still] hit the apple (1) Difference - the apple falls further before being hit or greater time [before being hit] (1) Award 2 marks for - The apple will be hit further down		2		2		
	(b)	(i)	$v_H = 3.0 \text{ [m s}^{-1}\text{]}$	1			1	1	
		(ii)	$v_v = 2.1 \text{ [m s}^{-1}\text{]}$ (from $3\tan 35^\circ$) (ecf on v_H) other methods possible (1) Attempt at use of $v^2 = u^2 + 2ax$ including allowing for incorrect use of signs (1) Correct substitution e.g. $0 = (2.1)^2 - 19.62h$ (1) (ecf on u_v) $h = 0.22 \text{ [m]}$ (1) Alternative: $v_v = 2.1 \text{ [m s}^{-1}\text{]}$ (from $3\tan 35^\circ$) (ecf on v_H) other methods possible (1) Attempt at use of $x = ut + \frac{1}{2}at^2$ including allowing for incorrect use of signs (1) Correct substitution: $h = 2.1 \times 0.2 - 4.9 \times 0.2^2$ (ecf on u_v) (1) $h = 0.224 \text{ [m]}$ (1)	1 1	1 1		4	4	